Evaluation runs: **reset** (s5-reset)

generated 2026-11-03 from brain-data 4f70fdcf4 2026-11-03 22:15:02 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): reset: expected violation; ctl: expected violation (the matched control is a violating one); twin: expected conforms. Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
1,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
1,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
2,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
2,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
2,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
3,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
4,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
5,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
5,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
5,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
6,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
6,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
6,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
7,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
7,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
7,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
8,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
8,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
8,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
9,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
9,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
9,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
10,1	1	1	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
10,3	6	6	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
10,4	24	24	conforms	0.0	2	1 h, 250 GB	1	1	brain03, 2026-11-02
10,5	120	120	conforms	0.1	2	1 h, 250 GB	1	1	brain01, 2026-11-02
10,6	720	720	conforms	0.6	2	1 h, 250 GB	1	1	brain01, 2026-11-02
10,7	5040	5040	conforms	4.3	2	1 h, 250 GB	1	1	brain01, 2026-11-02
10,8	40320	40320	conforms	36.3	10	1 h, 250 GB	1	1	brain01, 2026-11-02
10,9	362880	362880	conforms	336.4	10	1 h, 250 GB	1	1	brain01, 2026-11-02
10,10	3628800	3628800	conforms	3500.3	10	1 h, 250 GB	1	1	brain01, 2026-11-02

stateless: pair **ctl**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1,1	1	2	1	violation	0.0	2	1 h, 250 GB	2	2	brain03, 2026-11-02
1,3	6	2	1	violation	0.0	4	1 h, 250 GB	2	7	brain03, 2026-11-02
1,4	24	2	1	violation	0.0	2	1 h, 250 GB	2	25	brain03, 2026-11-02
2,1	1	4	3	violation	0.0	2	1 h, 250 GB	4	4	brain03, 2026-11-02
2,3	6	4	3	violation	0.0	2	1 h, 250 GB	4	19	brain03, 2026-11-02
2,4	24	4	3	violation	0.0	2	1 h, 250 GB	4	73	brain03, 2026-11-02
3,1	1	8	7	violation	0.0	2	1 h, 250 GB	8	8	brain03, 2026-11-02
3,3	6	8	7	violation	0.0	2	1 h, 250 GB	8	43	brain03, 2026-11-02
3,4	24	8	7	violation	0.1	2	1 h, 250 GB	8	169	brain03, 2026-11-02
4,1	1	16	15	violation	0.0	2	1 h, 250 GB	16	16	brain03, 2026-11-02
4,3	6	16	15	violation	0.0	2	1 h, 250 GB	16	91	brain03, 2026-11-02
4,4	24	16	15	violation	0.1	2	1 h, 250 GB	16	361	brain03, 2026-11-02
5,1	1	32	31	violation	0.0	2	1 h, 250 GB	32	32	brain03, 2026-11-02
5,3	6	32	31	violation	0.1	2	1 h, 250 GB	32	187	brain03, 2026-11-02
5,4	24	32	31	violation	0.1	2	1 h, 250 GB	32	745	brain03, 2026-11-02
6,1	1	64	63	violation	0.1	2	1 h, 250 GB	64	64	brain03, 2026-11-02
6,3	6	64	63	violation	0.1	2	1 h, 250 GB	64	379	brain03, 2026-11-02
6,4	24	64	63	violation	0.2	2	1 h, 250 GB	64	1513	brain03, 2026-11-02
7,1	1	128	127	violation	0.1	4	1 h, 250 GB	128	128	brain03, 2026-11-02
7,3	6	128	127	violation	0.2	4	1 h, 250 GB	128	763	brain03, 2026-11-02
7,4	24	128	127	violation	0.5	4	1 h, 250 GB	128	3049	brain03, 2026-11-02
8,1	1	256	255	violation	0.3	8	1 h, 250 GB	256	256	brain03, 2026-11-02
8,3	6	256	255	violation	0.5	8	1 h, 250 GB	256	1531	brain03, 2026-11-02
8,4	24	256	255	violation	1.1	6	1 h, 250 GB	256	6121	brain03, 2026-11-02
9,1	1	512	511	violation	0.7	16	1 h, 250 GB	512	512	brain03, 2026-11-02
9,3	6	512	511	violation	1.1	16	1 h, 250 GB	512	3067	brain03, 2026-11-02
9,4	24	512	511	violation	2.8	12	1 h, 250 GB	512	12265	brain03, 2026-11-02
10,1	1	1024	1023	violation	1.8	32	1 h, 250 GB	1024	1024	brain03, 2026-11-02
10,3	6	1024	1023	violation	2.8	30	1 h, 250 GB	1024	6139	brain03, 2026-11-02
10,4	24	1024	1023	violation	6.4	26	1 h, 250 GB	1024	24553	brain03, 2026-11-02
10,5	120	1024	1023	violation	27.4	30	1 h, 250 GB	1024	122761	brain01, 2026-11-02
10,6	720	1024	1023	violation	156.1	30	1 h, 250 GB	1024	736561	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1101.9	24	1 h, 250 GB	1024	5155921	brain01, 2026-11-02
10,8	40320	1024	1023	violation	9207.0	26	48 h, 250 GB	1024	41247361	brain01, 2026-11-03

stateless: pair **reset**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G^{expl}_{spec}	run
1,1	1	2	1	violation	0.0	2	1 h, 250 GB	2	2	brain03, 2026-11-02
1,3	6	2	1	violation	0.0	2	1 h, 250 GB	2	7	brain03, 2026-11-02
1,4	24	2	1	violation	0.0	2	1 h, 250 GB	2	25	brain03, 2026-11-02
2,1	1	4	3	violation	0.0	2	1 h, 250 GB	4	4	brain03, 2026-11-02
2,3	6	4	3	violation	0.0	2	1 h, 250 GB	4	19	brain03, 2026-11-02
2,4	24	4	3	violation	0.0	2	1 h, 250 GB	4	73	brain03, 2026-11-02
3,1	1	8	7	violation	0.0	2	1 h, 250 GB	8	8	brain03, 2026-11-02
3,3	6	8	7	violation	0.0	2	1 h, 250 GB	8	43	brain03, 2026-11-02
3,4	24	8	7	violation	0.0	2	1 h, 250 GB	8	169	brain03, 2026-11-02
4,1	1	16	15	violation	0.0	2	1 h, 250 GB	16	16	brain03, 2026-11-02
4,3	6	16	15	violation	0.0	2	1 h, 250 GB	16	91	brain03, 2026-11-02
4,4	24	16	15	violation	0.1	2	1 h, 250 GB	16	361	brain03, 2026-11-02
5,1	1	32	31	violation	0.0	2	1 h, 250 GB	32	32	brain03, 2026-11-02
5,3	6	32	31	violation	0.1	2	1 h, 250 GB	32	187	brain03, 2026-11-02
5,4	24	32	31	violation	0.1	2	1 h, 250 GB	32	745	brain03, 2026-11-02
6,1	1	64	63	violation	0.1	2	1 h, 250 GB	64	64	brain03, 2026-11-02
6,3	6	64	63	violation	0.1	2	1 h, 250 GB	64	379	brain03, 2026-11-02
6,4	24	64	63	violation	0.2	2	1 h, 250 GB	64	1513	brain03, 2026-11-02
7,1	1	128	127	violation	0.1	4	1 h, 250 GB	128	128	brain03, 2026-11-02
7,3	6	128	127	violation	0.3	6	1 h, 250 GB	128	763	brain03, 2026-11-02
7,4	24	128	127	violation	0.5	4	1 h, 250 GB	128	3049	brain03, 2026-11-02
8,1	1	256	255	violation	0.3	10	1 h, 250 GB	256	256	brain03, 2026-11-02
8,3	6	256	255	violation	0.5	8	1 h, 250 GB	256	1531	brain03, 2026-11-02
8,4	24	256	255	violation	1.1	8	1 h, 250 GB	256	6121	brain03, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{spec}^{expl}	run
9,1	1	512	511	violation	0.7	16	1 h, 250 GB	512	512	brain03, 2026-11-02
9,3	6	512	511	violation	1.0	18	1 h, 250 GB	512	3067	brain03, 2026-11-02
9,4	24	512	511	violation	2.4	16	1 h, 250 GB	512	12265	brain03, 2026-11-02
10,1	1	1024	1023	violation	1.2	30	1 h, 250 GB	1024	1024	brain03, 2026-11-02
10,3	6	1024	1023	violation	2.0	30	1 h, 250 GB	1024	6139	brain03, 2026-11-02
10,4	24	1024	1023	violation	4.6	34	1 h, 250 GB	1024	24553	brain03, 2026-11-02
10,5	120	1024	1023	violation	19.6	30	1 h, 250 GB	1024	122761	brain01, 2026-11-02
10,6	720	1024	1023	violation	113.2	34	1 h, 250 GB	1024	736561	brain01, 2026-11-02
10,7	5040	1024	1023	violation	801.4	30	1 h, 250 GB	1024	5155921	brain01, 2026-11-02
10,8	40320	1024	1023	violation	6572.0	30	48 h, 250 GB	1024	41247361	brain01, 2026-11-03

Not run (stateless). planned but never scheduled (grid points beyond the ladder): ctl 10 8 bag, twin 10 8 bag, reset 10 8 bag, reset 10 9 bag, ctl 10 9 bag, reset 10 10 bag, ctl 10 10 bag, twin 11 1 bag, reset 11 1 bag, ctl 11 1 bag, reset 11 3 bag, twin 11 3 bag, ctl 11 3 bag, ctl 11 4 bag, reset 11 4 bag, twin 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, reset 12 3 bag, twin 12 3 bag, ctl 12 3 bag, twin 12 4 bag, reset 12 4 bag, … (37 instances in all).

left out by the table rules (invalid: foreign load): reset 20 11 bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
9,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
9,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
9,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
10,1	1	1	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
10,3	6	6	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
10,4	24	24	conforms	0.0	4	1 h, 250 GB	brain02, 2026-11-02
10,5	120	120	conforms	0.1	2	1 h, 250 GB	brain01, 2026-11-02
10,6	720	720	conforms	0.3	2	1 h, 250 GB	brain01, 2026-11-02
10,7	5040	5040	conforms	2.1	2	1 h, 250 GB	brain01, 2026-11-02
10,8	40320	40320	conforms	17.1	10	1 h, 250 GB	brain01, 2026-11-02
10,9	362880	362880	conforms	157.8	10	1 h, 250 GB	brain01, 2026-11-02
10,10	3628800	3628800	conforms	1601.8	10	1 h, 250 GB	brain01, 2026-11-02

stateful: pair **ctl**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,1	1	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,3	6	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,4	24	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,1	1	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	6	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	24	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,1	1	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,3	6	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,4	24	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,1	1	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,3	6	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,4	24	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,1	1	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,3	6	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,4	24	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,1	1	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,3	6	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,4	24	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,1	1	128	127	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,3	6	128	127	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,4	24	128	127	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
8,1	1	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02
8,3	6	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02
8,4	24	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02
9,1	1	512	511	violation	0.2	14	1 h, 250 GB	brain02, 2026-11-02
9,3	6	512	511	violation	0.3	14	1 h, 250 GB	brain02, 2026-11-02
9,4	24	512	511	violation	0.3	14	1 h, 250 GB	brain02, 2026-11-02
10,1	1	1024	1023	violation	0.8	34	1 h, 250 GB	brain02, 2026-11-02
10,3	6	1024	1023	violation	0.8	36	1 h, 250 GB	brain02, 2026-11-02
10,4	24	1024	1023	violation	0.8	36	1 h, 250 GB	brain02, 2026-11-02
10,5	120	1024	1023	violation	0.8	34	1 h, 250 GB	brain01, 2026-11-02
10,6	720	1024	1023	violation	0.9	34	1 h, 250 GB	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1.8	34	1 h, 250 GB	brain01, 2026-11-02
10,8	40320	1024	1023	violation	9.0	34	1 h, 250 GB	brain01, 2026-11-02
10,9	362880	1024	1023	violation	77.5	34	1 h, 250 GB	brain01, 2026-11-02
10,10	3628800	1024	1023	violation	791.0	34	1 h, 250 GB	brain01, 2026-11-02

stateful: pair **reset**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1,1	1	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,3	6	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
1,4	24	2	1	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,1	1	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,3	6	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
2,4	24	4	3	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,1	1	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,3	6	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
3,4	24	8	7	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,1	1	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,3	6	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
4,4	24	16	15	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,1	1	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,3	6	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
5,4	24	32	31	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,1	1	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,3	6	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
6,4	24	64	63	violation	0.0	4	1 h, 250 GB	brain02, 2026-11-02
7,1	1	128	127	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
7,3	6	128	127	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
7,4	24	128	127	violation	0.1	4	1 h, 250 GB	brain02, 2026-11-02
8,1	1	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02
8,3	6	256	255	violation	0.1	6	1 h, 250 GB	brain02, 2026-11-02
8,4	24	256	255	violation	0.1	8	1 h, 250 GB	brain02, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
9,1	1	512	511	violation	0.2	14	1 h, 250 GB	brain02, 2026-11-02
9,3	6	512	511	violation	0.2	18	1 h, 250 GB	brain02, 2026-11-02
9,4	24	512	511	violation	0.2	16	1 h, 250 GB	brain02, 2026-11-02
10,1	1	1024	1023	violation	0.5	38	1 h, 250 GB	brain02, 2026-11-02
10,3	6	1024	1023	violation	0.5	36	1 h, 250 GB	brain02, 2026-11-02
10,4	24	1024	1023	violation	0.5	36	1 h, 250 GB	brain02, 2026-11-02
10,5	120	1024	1023	violation	0.6	34	1 h, 250 GB	brain01, 2026-11-02
10,6	720	1024	1023	violation	0.6	36	1 h, 250 GB	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1.6	32	1 h, 250 GB	brain01, 2026-11-02
10,8	40320	1024	1023	violation	9.0	34	1 h, 250 GB	brain01, 2026-11-02
10,9	362880	1024	1023	violation	79.8	32	1 h, 250 GB	brain01, 2026-11-02
10,10	3628800	1024	1023	violation	807.5	32	1 h, 250 GB	brain01, 2026-11-02

Not run (stateful). planned but never scheduled (grid points beyond the ladder): ctl 10 8 bag, twin 10 8 bag, reset 10 8 bag, twin 11 1 bag, reset 11 1 bag, ctl 11 1 bag, reset 11 3 bag, twin 11 3 bag, ctl 11 3 bag, ctl 11 4 bag, reset 11 4 bag, twin 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, reset 12 3 bag, twin 12 3 bag, ctl 12 3 bag, twin 12 4 bag, reset 12 4 bag, ctl 12 4 bag, ctl 13 1 bag, reset 13 1 bag, twin 13 1 bag, … (33 instances in all).

left out by the table rules (invalid: foreign load): reset 20 11 bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1	1	1	conforms	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
1,3	6	6	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
1,4	24	24	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
2,1	1	1	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
2,3	6	6	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
2,4	24	24	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
3,1	1	1	conforms	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
3,3	6	6	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,4	24	24	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
4,1	1	1	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
4,3	6	6	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
4,4	24	24	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
5,1	1	1	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
5,3	6	6	conforms	0.2	4	10 min, 16 GB	0.0	0.2	0.0	in-memory	brain01, 2026-11-02
5,4	24	24	conforms	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
6,1	1	1	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
6,3	6	6	conforms	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
6,4	24	24	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7,1	1	1	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
7,3	6	6	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7,4	24	24	conforms	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
8,1	1	1	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,3	6	6	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,4	24	24	conforms	0.4	4	10 min, 16 GB	0.4	0.1	0.0	in-memory	brain01, 2026-11-02
9,1	1	1	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
9,3	6	6	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
9,4	24	24	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
10,1	1	1	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
10,3	6	6	conforms	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
10,4	24	24	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
10,5	120	120	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
10,6	720	720	conforms	0.3	4	10 min, 16 GB	0.3	0.2	0.0	in-memory	brain01, 2026-11-02
10,7	5040	5040	conforms	1.4	4	10 min, 16 GB	1.4	1.3	0.0	in-memory	brain01, 2026-11-02
10,8	40320	40320	conforms	10.7	10	10 min, 16 GB	10.7	10.7	0.0	in-memory	brain01, 2026-11-02
10,9	362880	362880	conforms	98.5	9	1 h, 250 GB	98.5	97.3	0.0	merge	brain01, 2026-11-03
10,10	3628800	3628800	conforms	996.2	10	1 h, 250 GB	992.2	996.2	0.0	merge	brain01, 2026-11-03

offline: pair **ctl**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1	1	2	1	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
1,3	6	2	1	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
1,4	24	2	1	violation	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
2,1	1	4	3	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
2,3	6	4	3	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
2,4	24	4	3	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
3,1	1	8	7	violation	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
3,3	6	8	7	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
3,4	24	8	7	violation	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
4,1	1	16	15	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
4,3	6	16	15	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
4,4	24	16	15	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
5,1	1	32	31	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
5,3	6	32	31	violation	0.2	4	10 min, 16 GB	0.0	0.2	0.0	in-memory	brain01, 2026-11-02
5,4	24	32	31	violation	0.3	4	10 min, 16 GB	0.2	0.3	0.0	in-memory	brain01, 2026-11-02
6,1	1	64	63	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
6,3	6	64	63	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
6,4	24	64	63	violation	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
7,1	1	128	127	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
7,3	6	128	127	violation	0.2	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7,4	24	128	127	violation	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
8,1	1	256	255	violation	0.1	6	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,3	6	256	255	violation	0.1	6	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,4	24	256	255	violation	0.4	6	10 min, 16 GB	0.4	0.1	0.0	in-memory	brain01, 2026-11-02
9,1	1	512	511	violation	0.2	12	10 min, 16 GB	0.1	0.1	0.1	in-memory	brain01, 2026-11-02
9,3	6	512	511	violation	0.3	12	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
9,4	24	512	511	violation	0.2	12	10 min, 16 GB	0.1	0.1	0.1	in-memory	brain01, 2026-11-02
10,1	1	1024	1023	violation	0.4	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
10,3	6	1024	1023	violation	0.4	24	10 min, 16 GB	0.2	0.3	0.1	in-memory	brain01, 2026-11-02
10,4	24	1024	1023	violation	0.4	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
10,5	120	1024	1023	violation	0.5	24	10 min, 16 GB	0.1	0.4	0.1	in-memory	brain01, 2026-11-02
10,6	720	1024	1023	violation	0.5	24	10 min, 16 GB	0.3	0.4	0.1	in-memory	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1.5	24	10 min, 16 GB	1.4	0.3	0.1	in-memory	brain01, 2026-11-02
10,8	40320	1024	1023	violation	10.8	9	1 h, 250 GB	10.7	0.3	0.1	merge	brain01, 2026-11-03

offline: pair **reset**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1,1	1	2	1	violation	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
1,3	6	2	1	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
1,4	24	2	1	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
2,1	1	4	3	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
2,3	6	4	3	violation	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
2,4	24	4	3	violation	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
3,1	1	8	7	violation	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain01, 2026-11-02
3,3	6	8	7	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
3,4	24	8	7	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
4,1	1	16	15	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain01, 2026-11-02
4,3	6	16	15	violation	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
4,4	24	16	15	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
5,1	1	32	31	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
5,3	6	32	31	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
5,4	24	32	31	violation	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
6,1	1	64	63	violation	0.2	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
6,3	6	64	63	violation	0.2	4	10 min, 16 GB	0.1	0.2	0.0	in-memory	brain01, 2026-11-02
6,4	24	64	63	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7,1	1	128	127	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain01, 2026-11-02
7,3	6	128	127	violation	0.2	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
7,4	24	128	127	violation	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain01, 2026-11-02
8,1	1	256	255	violation	0.2	6	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,3	6	256	255	violation	0.1	6	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain01, 2026-11-02
8,4	24	256	255	violation	0.4	6	10 min, 16 GB	0.4	0.2	0.0	in-memory	brain01, 2026-11-02
9,1	1	512	511	violation	0.2	12	10 min, 16 GB	0.1	0.2	0.1	in-memory	brain01, 2026-11-02
9,3	6	512	511	violation	0.3	12	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
9,4	24	512	511	violation	0.3	12	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
10,1	1	1024	1023	violation	0.5	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
10,3	6	1024	1023	violation	0.5	24	10 min, 16 GB	0.2	0.3	0.1	in-memory	brain01, 2026-11-02
10,4	24	1024	1023	violation	0.4	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
10,5	120	1024	1023	violation	0.4	24	10 min, 16 GB	0.1	0.3	0.1	in-memory	brain01, 2026-11-02
10,6	720	1024	1023	violation	0.4	24	10 min, 16 GB	0.3	0.3	0.1	in-memory	brain01, 2026-11-02
10,7	5040	1024	1023	violation	1.5	24	10 min, 16 GB	1.4	0.3	0.1	in-memory	brain01, 2026-11-02
10,8	40320	1024	1023	violation	11.2	9	1 h, 250 GB	10.7	0.3	0.5	merge	brain01, 2026-11-03

Not run (offline). planned but never scheduled (grid points beyond the ladder): twin 11 1 bag, reset 11 1 bag, ctl 11 1 bag, reset 11 3 bag, twin 11 3 bag, ctl 11 3 bag, ctl 11 4 bag, reset 11 4 bag, twin 11 4 bag, ctl 12 1 bag, reset 12 1 bag, twin 12 1 bag, reset 12 3 bag, twin 12 3 bag, ctl 12 3 bag, twin 12 4 bag, reset 12 4 bag, ctl 12 4 bag, ctl 13 1 bag, reset 13 1 bag, twin 13 1 bag, reset 13 3 bag, ctl 13 3 bag, twin 13 3 bag, ... (27 instances in all).

refused (an input missing at run time): ctl 10 9 bag, reset 10 9 bag, reset 10 10 bag, ctl 10 10 bag, reset 20 11 bag.